

Test Program 1

Image Taken (ESP32-S3-EYE) Actual Size + Zoomed:



Zoomed:



Image Taken Via iPhone:



Image Comparison Paragraph

A picture of the same object taken with an iPhone is clear, bright, and shows the subject and surroundings with natural colors and good detail. The image from the board's camera is very dark, has a green tint, and appears pixelated due to the limited resolution and memory constraints. Small details are difficult to see, and overall quality is lower, showing differences in brightness, saturation, and resolution. These differences suggest that applying brightness or color corrections, or other augmentations, may help make the board images more similar to higher-quality camera outputs for training or analysis.

Test Program 2

Program Classification Output

```
Input t=== START ===  
Model initialized!  
Input1 raw output: -12 -55 0 31 -9 23 31 -31 1 -42  
Input1 predicted class: 3  
Inference time (us): 143976  
  
Input2 raw output: 30 47 -31 -34 -42 -69 -67 -67 55 1  
Input2 predicted class: 8  
Inference time (us): 143760
```

Python Classification Output

```
img1 = test_images[0]  
img2 = test_images[1]  
  
label1 = test_labels[0]  
label2 = test_labels[1]  
  
print("Input1 true label:", label1)  
print("Input2 true label:", label2)
```

✓ 0.0s

```
Input1 true label: 3  
Input2 true label: 8
```

Comparison Table

Input #	Actual Label	VsCode Output	Tflite Output	Embedded Output	Inference Time (us)
1	3	3	3	-12 -55 0 31 -9 23 31 -31 1 -42	143976
2	8	8	8	30 47 -31 -34 -42 -69 -67 -67 55 1	143760